USN | | | | | | | | | | [22EE32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

ELECTRONIC CIRCUITS AND APPLICATIONS

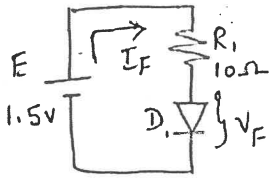
Time: 3 Hours **Maximum Marks: 100**

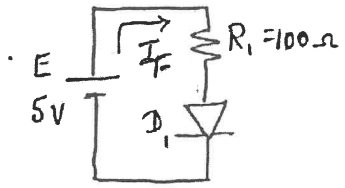
Instructions: **i.** Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **5 X 2=10**

1.	**a.**	Mention the effect of temperature on diode characteristics.	1	CO1	**02**
	b.	Define Transformer utilization factor.	1	CO2	**02**
	c.	Draw the structure of BJT and explain in brief	2	CO3	**02**
	d.	Mention the advantages and Disadvantages of Power MOSFET.	1	CO4	**02**
	e.	State the Barkhausen criterion for an oscillator.	1	CO6	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the classification of material based on conductivity and energy band diagram	2	CO1	**05**
		OR			
	b.	Outline the charge carrier diffusion phenomenon across a PN junction. Explain the effect of forward and reverse biasing on the depletion region.	2	CO1	**05**
3.	**a.**	Compare the performance of a half-wave rectifier and a full-wave rectifier.	2	CO2	**05**
		OR			
	b.	Explain the operation of a full wave rectifier with help of diagram.	2	CO2	**05**
4.	**a.**	Explain the NPN transistor operation with the help of a neat diagram.	2	CO3	**05**
		OR			
	b.	Sketch and explain typical BJT common-emitter input and output characteristics.	2	CO3	**05**
5.	**a.**	Discuss the working of class B power amplifiers with necessary diagrams.	2	CO4	**05**
		OR			
	b.	Construct the circuit diagram of Buck-Boost Converters and explain the same.	3	CO4	**05**
6.	**a.**	Illustrate the circuit of a RC Phase shift oscillator and discuss its working.	2	CO5	**05**
		OR			

	b.	Describe the working of a Colpitt's Oscillator.	2	CO5	**05**
7.	**a.**	Calculate I_F for the diode circuit given and assume diode has $V_F = 0.7V$ and $r_d = 0$ then recalculate the current taking $r_d = 0.25\Omega$. 	3	CO6	**05**
		OR			
	b.	For the half wave rectifier, source is a sinusoid of 120V rms at a frequency of 600Hz. The load resistor is 5Ω. Determine **i.** The average load current. **ii.** The average power absorbed by the load.	2	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Draw the circuit diagram of a positive clipper circuit and explain its working with necessary input and output waveform.	2	CO1	**10**
		OR			
	b.	Construct and explain the working of clamper circuit with necessary input and output waveform.	2	CO1	**10**
9.	**a.**	Sketch the circuit of a half-wave rectifier and explain its operation. Derive Vdc, Vrms& PIV for it.	2	CO2	**10**
		OR			
	b.	Construct the circuit and explain in brief about C, LC and π Filter.	3	CO2	**10**
10.	**a.**	Sketch a block diagram representation for an n-channel JFET and explain its operation.	2	CO3	**10**
		OR			
	b.	Draw the cross section diagram for an enhancement-mode MOSFET and explain its operation.	3	CO3	**10**
11.	**a.**	Discuss the class C power amplifiers with necessary diagrams.	2	CO4	**10**
		OR			
	b.	Explain the basic operation of Buck converter and derive the expression for duty cycle, D	2	CO4	**10**
12.	**a.**	Derive the frequency of oscillations in Wein bridge oscillator by clearly explaining the operation of the same	4	CO5	**10**
		OR			
	b.	Explain Hartley oscillator and derive the equation for oscillation?	2	CO5	**10**

13.	**a.**	Differentiate between Zener breakdown and Avalanche breakdown. Calculate the diode current for the given circuit and draw the dc load line for the circuit on the diode forward characteristic. 	2	CO6	**10**
		OR			
	b.	Compare FET and MOSFET. A BJT has the following measured current levels: $I_C = 5.25mA$ and $I_B = 100\mu A$. Calculate I_E, α_{dc} and β_{dc} and determine the new I_B level to give $I_C = 15mA$.	2	CO6	**10**